EXPERIMENT 4:

Verification of System Properties: Linearity, Time Invariance, Causality and Stability

Objective :

To verify the linearity and time-invariance properties of continuous-time and discrete-time systems using suitable input signals and system responses.

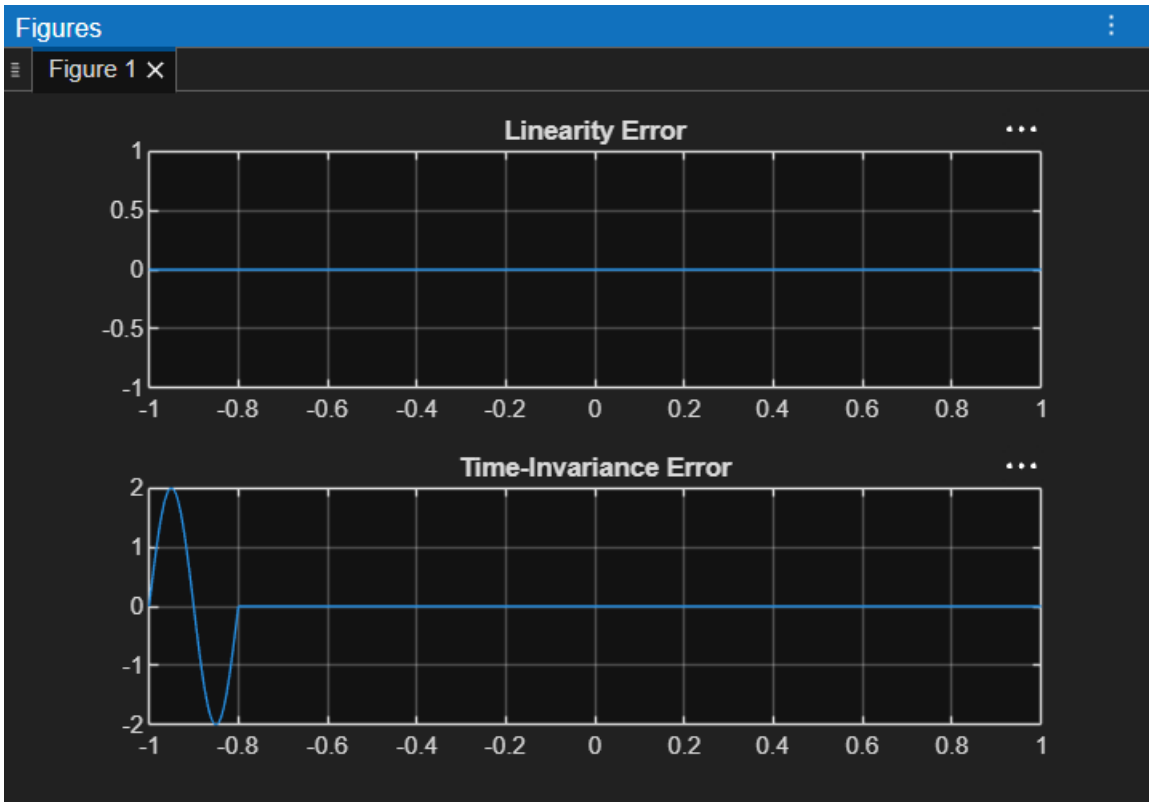
Procedure

1. Select a representative CT or DT system.
2. Generate two independent test inputs.
3. Compute the individual system outputs.
4. Form a combined input and compare its system output with the corresponding linear
5. combination of outputs.
6. Delay the input by t0 and compare the system response with the correspondingly
7. delayed original output.
8. Calculate numerical differences and classify the system

MATLAB Coding

```
% Experiment 4: Linearity and Time Invariance
clc; clear; close all;
t = -1:0.001:1;
x1 = sin(2*pi*5*t); x2 = cos(2*pi*5*t);
a = 2; b = 3; t0 = 0.2;
T = @(x) 2*x;
lhs = T(a*x1+b*x2);
rhs = a*T(x1)+b*T(x2);
linError = max(abs(lhs-rhs));
xDelay = sin(2*pi*5*(t-t0));
lhsTI = T(xDelay);
rhsTI = T(x1);
rhsTIshift = interp1(t,rhsTI,t-t0,'linear',0);
tiError = max(abs(lhsTI-rhsTIshift));
figure;
subplot(2,1,1); plot(t,lhs-rhs); grid on; title('Linearity Error');
subplot(2,1,2); plot(t,lhsTI-rhsTIshift); grid on; title('Time-Invariance Error');
fprintf('Linearity error = %g\n',linError);
fprintf('Time-invariance error = %g\n',tiError);
```

OUTPUT:



RESULT:

The discrete-time system $y(n)=x(n)+x(n-1)$ was implemented using MATLAB. The output sequence was obtained by combining the present and previous input samples, and the input and output signals were displayed using stem plots.

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage	Marks(10)	Marks obtained
1. Generate and visualize basic continuous-time signals such as unit step, unit impulse, ramp, sinusoidal and exponential.	**CT Signal Generation & MATLAB Implementation**	Generates all five CT signals accurately with a suitable time vector and step size. Uses correct MATLAB commands . Plots are correctly scaled, with titles, axis labels and grid.	Generates the required CT signals and plots correctly, with minor errors in code, time range, parameters or labelling.	Generates basic CT signals but has noticeable errors in code, step or impulse representation, or plots, and needs assistance.	Unable to generate the required CT signals or produce meaningful plots.	**25**	**2.5**	
2. Generate and visualize basic discrete-time signals such as unit sample, unit step, ramp, sinusoidal and exponential sequences.	**DT Signal Generation & MATLAB Implementation**	Generates all five DT sequences accurately with a proper index vector n. Uses stem correctly. Plots are correctly scaled, with titles, axis labels and grid.	Generates the required DT sequences and plots correctly, with minor errors in index range, stem usage, parameters or labelling.	Generates basic DT sequences but has noticeable errors in code, indexing or stem plots, and needs assistance.	Unable to generate the required DT sequences or produce meaningful stem plots.	**25**	**2.5**	
3. Classify CT and DT signals as periodic or aperiodic and identify their waveform characteristics.	**Signal Classification & Periodicity Analysis**	Correctly classifies all signals using DT. Determines the period where applicable. Identifies frequency, phase, and growth or decay. Explains the	Classifies most signals correctly and finds most periods, with minor errors or limited justification.	Gives basic classifications . Period calculation or justification is incomplete and needs guidance.	Unable to classify the signals correctly or justify the classification .	**25**	**2.5**	

		reasoning clearly with mathematical conditions and waveform evidence.						
4. Analyze the amplitude and time-index characteristics of discrete-time signals.	**Amplitude & Time-Index Analysis**	Correctly determines maximum and minimum amplitude, the index n at which each occurs, and signal energy. Correctly shows the effect of time shifting and folding on the index. Interprets the results clearly.	Analyzes most amplitude and index characteristics correctly, with minor errors in values or interpretation.	Gives basic amplitude and index analysis with noticeable errors, and needs guidance.	Unable to analyze the amplitude and time-index characteristics or interpret the results.	**25**	**2.5**	
					Total	**100**	**10**	